

Growth Factors and their receptors in cancer metastases.

Metastatic, rather than primary tumours are responsible for ninety percent cancer deaths. Despite significant advances in the understanding of molecular and cellular mechanisms in tumour metastases, there are limitations in preventive treatment of metastatic tumours. Much evidence arising from laboratory and clinical studies suggests that growth factors and their receptors are implicated in cancer metastases development. We review the origin and production of growth factors and their receptors in all stages of cancer metastases including epithelial-mesenchymal transition, cancer cell invasion and migration, survival within the circulation, seeding at distant organs and metastatic tumour angiogenesis. The functions of growth factors and their receptors are also discussed. This review presents the efforts made in understanding this challenge to aid in the development of new treatment strategies for cancer metastases.